H.5. Models and probing

I. Additional experimental results

In this section we provide additional experimental results discussed in the main text. We used (Rubinstein & Uselis, 2025) for conducting the experiments.

I.1. Linearity and compositional generalization

We provide an extended view of the relationship between projected linearity (R^2) and compositional generalization under different train/test splits. We follow the same pipeline as in Section 5.2: for each dataset/model, we train one linear probe per concept and evaluate average compositional accuracy on held-out concept combinations. The full split-ablation results are shown in Fig. 23: across held-out fractions, higher projected R^2 is still associated with higher held-out accuracy, while the trend flattens in near-ceiling regimes (especially on dSprites and MPI3D).

For each split level, we set a test fraction $\rho_{\text{test}} \in \{0.95, 0.90, 0.50, 0.30, 0.10\}$ and train on the remaining $1 - \rho_{\text{test}}$ fraction. For every model/split setting, probes are trained with Adam for 1000 epochs using cosine decay to zero, and we select the best result over learning rates $\{10^{-3}, 10^{-2}\}$.

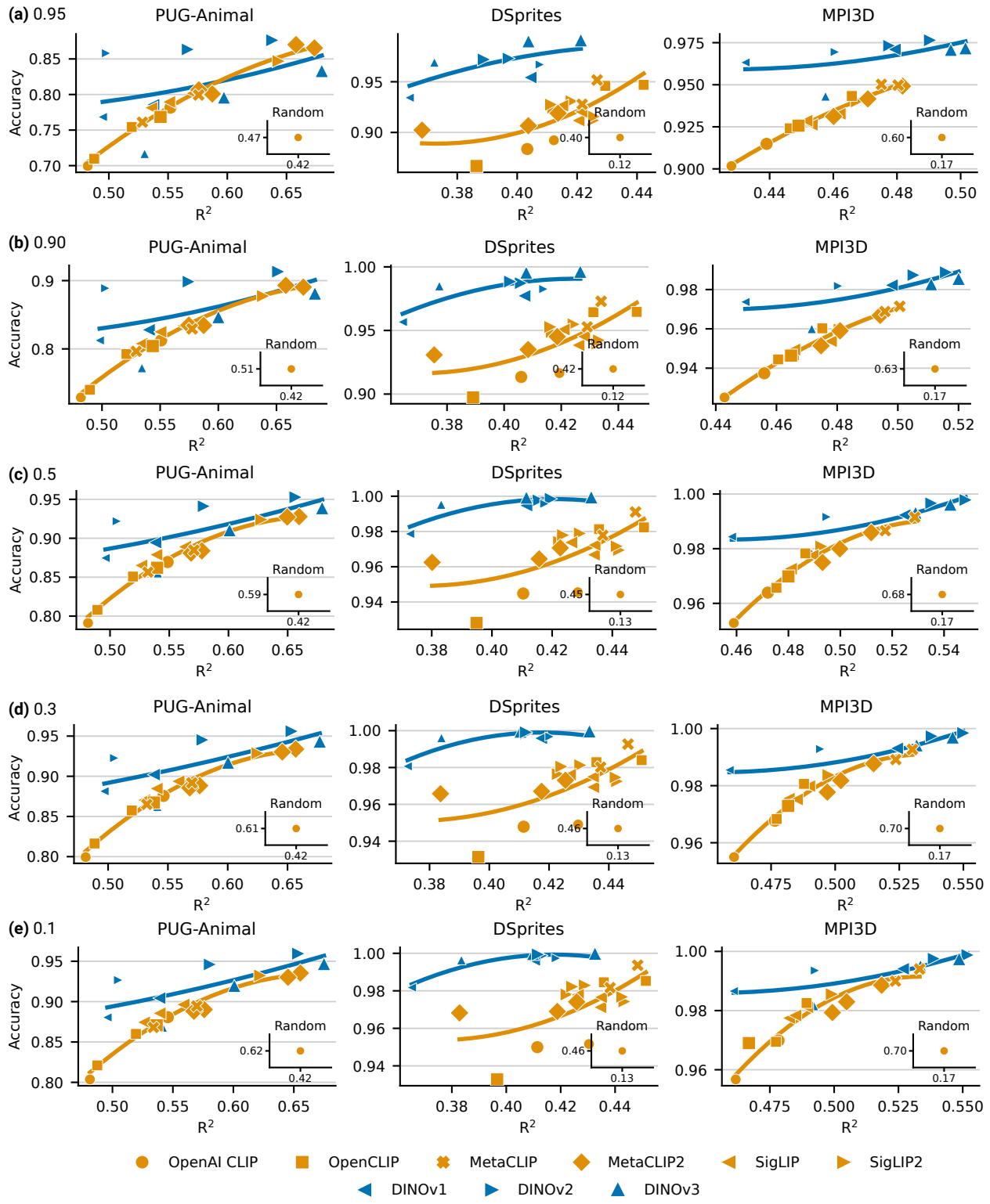

Figure 23. **Projected R^2 vs compositional accuracy across split regimes.** Columns correspond to datasets (PUG-Animal, dSprites, MPI3D). Rows **(a)–(e)** correspond to different held-out test fractions ρ_{test} (shown next to each row). The x-axis is projected R^2 and the y-axis is compositional accuracy on held-out combinations. Each marker is a model checkpoint (legend); the inset in each panel reports the randomly initialized OpenCLIP baseline for the same dataset/split. Across split regimes, higher projected R^2 is consistently associated with higher held-out compositional accuracy.

I.2. Orthogonality of factors

Setup. For each dataset/model, we extract image embeddings $\boldsymbol{z_c}$ and recover concept factors $\{\boldsymbol{u}_{i,v}\}$ as in Section 5.1. For orthogonality, we compute all quantities in the original embedding space (without projection by $P_{\boldsymbol{W}}$). For each concept i, we define centered and normalized factor directions:

$$\bar{\boldsymbol{u}}_i := \frac{1}{|\mathcal{C}_i|} \sum_{v \in \mathcal{C}_i} \boldsymbol{u}_{i,v}, \tag{97}$$

$$\boldsymbol{d}_{i,v} := \boldsymbol{u}_{i,v} - \bar{\boldsymbol{u}}_i, \qquad \tilde{\boldsymbol{d}}_{i,v} := \frac{\boldsymbol{d}_{i,v}}{\|\boldsymbol{d}_{i,v}\|}. \tag{98}$$

Metrics. We measure orthogonality via absolute cosine between direction vectors (lower $|\cos| \Rightarrow$ greater orthogonality). For any concepts $i \neq j$, we define

$$\mathrm{Orth}(i,j) := \frac{1}{|\mathcal{C}_i||\mathcal{C}_j|} \sum_{a \in \mathcal{C}_i} \sum_{b \in \mathcal{C}_j} \left|\tilde{\boldsymbol{d}}_{i,a}^{\intercal} \tilde{\boldsymbol{d}}_{j,b}\right|,$$

$$\mathrm{Orth}(i,i) := \frac{1}{|\mathcal{C}_i|(|\mathcal{C}_i|-1)} \sum_{\substack{a,b \in \mathcal{C}_i \\ a \neq b}} \left|\tilde{\boldsymbol{d}}_{i,a}^{\intercal} \tilde{\boldsymbol{d}}_{i,b}\right|.$$

We report $\mathrm{Orth}(i,i)$ as *within-concept direction similarity* and $\mathrm{Orth}(i,j)$ (for $i \neq j$) as *across-concept direction similarity*. In this parameterization, stronger cross-concept orthogonality corresponds to lower $\mathrm{Orth}(i,j)$.

Results. In Fig. 24, we provide the full per-model/per-dataset orthogonality matrices (including a randomly-initialized baseline) across the three datasets used in the main text.